

Developing a Personalized Multi-Dimensional Framework using Business Intelligence Techniques in Banking

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ABSTRACT

Intelligent techniques have been used in the marketing and sales sectors of business to improve analysis, increase revenues and save time. In customer-centric institutions, one of the areas in which intelligent techniques and data mining algorithms have been used is the personalization for enhanced CRM (Customer Relationship Management) performance. However, with a growing number of customers, the diversity of products on offer, the complex behavior of customer groups and the continuous change of personalization parameters, producing a tailored personalized recommendation that predicts their future needs is a challenging task.

In this paper, we propose multi dimension personalization framework architecture to improve personalized targeting. The framework presented improves on the automation of existing systems by using multiple supervised and unsupervised data mining techniques, and enhances the level of targeting by considering more effective dimensions in multiple stages of the framework. A theoretical case study explaining the practical working and perceived advantages of the new framework is presented.

CCS Concepts

• **Applied computing** → **Enterprise computing**; **Business process management**; **Business intelligence**

Keywords

Association Rules; Business Intelligence; Clustering; Customer churn; Dimensional Modeling; Group Profile; Intelligent DSS; Object Profiling; Personalization; Recommendation; User Profile; CRM.

1. INTRODUCTION

Personalization is the ability to provide items, services or actions that are tailored to individuals' needs in the right time. Based on analysis of knowledge about their preferences and past behavior,

a personalization technique uses information generated to predict customers' next activity or to build an intelligent decision support system. Personalization is a focus point of researchers in various business domains; like banking, telecom, education and others, as well as for data scientists. Building a personalized recommendation engine is considered a challenge in both computer science and CRM.

The personalization process involves the use of continuously refined profiles of current and potential customers in order to anticipate and respond to their needs. It draws on a combination of business process and intelligent techniques to discover the knowledge about the customers and answer questions like, "who are the customers?", "what do they do?" and "what do they like?" Answering these questions is central to any business in their quest to find competitive advantages to acquire new customers, and to develop and retain the existing ones. In addition, to evade or at least decrease the potential time, resources and cost losses resulting from miss targeting and customer churn.

Intelligent CRM is achieved through the incorporation of data mining techniques within the CRM cycle. Organizations are increasingly recognizing the benefits that can be realized from the use of data mining and are adopting it as a part of their decision making process. However, as pointed out by Gunjan Mansingh, organizations must also recognize that these benefits will only be realized if there is a clear understanding of the objective of the data mining process and of the data itself [1].

The combination of applying business intelligence techniques in CRM to produce a personalized model leads to better planning, more accurate decision support, product tailoring and service innovation (web-sites tailored to customer needs, the development of mass customization), providing a 360 degree view of the customer, calculating the CLTV (Customer lifetime value) of the customer, developing personalized actions, cross-selling/up-selling various products/services to customers, predicting the-next-best-activity, recommendation in the right time to increase the customer satisfaction and loyalty and switching from a generic campaign and targeting campaign to a one-to-one Campaign [2].

This paper proposes a novel personalization framework that combines two existing approaches and embeds data mining techniques to be the base of generating next best tailored activities that takes in consideration multiple sets of personalization dimensions.

The paper is organized as follow. Section 2 explores related work done in this area, Section 3 describes the MCMD (Multi-Criteria Multi-Dimensions) personalization framework, and Section 4

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INFOS '16, May 09-11, 2016, Giza, Egypt

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DOI: <http://dx.doi.org/10.1145/2908446.2908488>

explains the framework through a sample theoretical case study. The discussion and conclusion are presented in Section 5.

2. Related work

Personalization aims to deliver a tailored action or service to the right customer. There are many personalization approaches in the literature, each has different steps and leads to different results. Selecting a specific approach is done based on content availability and business needs. The main personalization approaches are described below.

2.1 Business-Driven Approach (Top-down)

In this approach, the decision of what actions to deliver is made before the definition of customers' profiles. It is similar to what happens in direct marketing campaigns, where top management decides which customers to approach with a product based on their business sense. The top-down approaches main drawback is coverage; as they do not cover the whole customer base. Although it may satisfy the effectiveness of targeting factor, for example by reducing customers churn in the segment targeted. In this approach the supervised data mining techniques are applied for the classification process. The limitations are in the coverage, and effectiveness of targeting. Also, because of the complex behavior, the customer's decision of leaving or remaining loyal, as the decision only depends on the offered products [3].

2.2 Knowledge Discovery Approach (Bottom-Up)

Knowledge discovery approaches are more of Bottom-up methods and involve the use of front office personnel. Techniques in this category are implemented in two steps: 1) build customer profiles; and 2) decide the proper action. They cannot be fully automated, so these approaches are typically very time consuming and may not always be efficient. Their advantage is that targeting can be very effective because each profile is thoroughly analyzed before generating a proper action. For that reason, it is suitable for financial advisors working with a small number of wealthy clients, although, the success will still be based on the dimensions considered in crafting the profile.

An example of this approach is in the work done in [4] on mobile user's data. The authors applied knowledge discovery approach for the usage and multidimensional analysis using Apriori mining algorithm considering time and location to provide tailored offers for each segment based on the discovered preferences. The big limitation they had was the automation of the process. Another limitation may be present if not all mobile users allow the public sharing of personal information features, which will decrease the level of targeting.

2.3 Dimensions of Personalization

There are many dimensions that could be considered for segmentation to improve the personalized targeting and ultimately get the one-to-one customer scoring. Categories of these dimensions are:

Socio-demographics Segmentation Dimension: describe the social positioning of a customer, and their geographical location (age, gender, income, employment, and city or region). Since customers sharing the same value for these attributes (think age or gender) tends to be interested in the same products. Residence and region are used because they may indicate the culture of the customer towards a certain activity [5] [6] [7]. This dimension,

although in general very indicative, could sometimes be misleading as it fails to consider practical behavior of usage which is different according to everyone.

Behavioral Segmentation Dimension includes the Usage, Payment, and Frequency of the customers towards the service or product [8]. It hence provides segmentation based on the behavior. However, usage may be seasonal or one-off-purchase, hence using the behavioral segmentation in isolation of the socio-demographic dimensions could lead to misclassification or to classification that is only correct for a short period of time.

Life-Stages Segmentation Dimension Life-stages present opportunities for promoting products and services that address the particular needs of customers in different stages of their life. The organization recognizes the important life-stage events and associates them to consuming behaviors [9]. However, this dimension follows a static implementation, and is not quickly responsive to changes.

Customer lifetime value Dimension CLTV represents the present value of the future cash flows attributed to the customer during his/her entire relationship with the company. It is used to predict the most profitable group of customers [10] [11]. CLTV focuses only on the revenue from the segments targeted, hence will not be able to predict the next activity for the customers identified since it misses other critical segmentation dimensions.

Considering any single set of these dimensions on its own leads to loss in the level of personalization accuracy. On the other hand, considering all the dimensions as inputs, makes the problem extremely complex to be modeled in a single framework.

3. MDMC Personalization Framework

The proposed framework is a hybrid approach that aims to enhance the traditional personalization frameworks drawbacks. The framework goal is to provide control over the Decision-Making process, enhance automation of the personalization process, and improve effectiveness of targeting and coverage of customers by considering multiple sets of dimensions over multiple stages of the framework with an order defined between the considered dimensions.

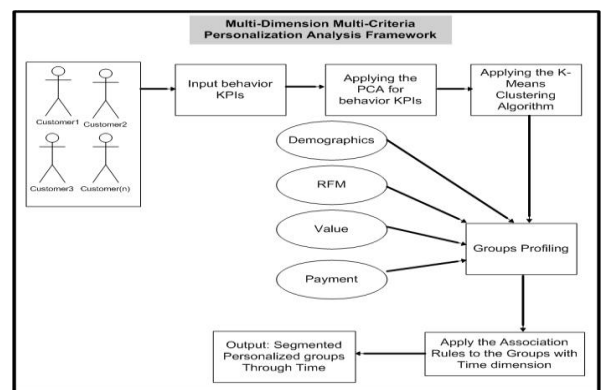


Figure 1: Multi-Dimensions Multi-Criteria Personalization Framework

The issue of automating the profiling is important to keep the cost low and the CRM processes speed high. We propose to solve it in the framework by applying unsupervised data mining technique in knowledge data discovery process followed by a

supervised classification technique for auto profiling process getting its rules from the first run after applying the business profiling. The automation leads to reach more enhancements and increase the efficiency of the new framework by reducing the implementation time and decreasing the human errors.

The MDMC framework also aims to satisfy a high level of targeting measure while maintaining a high coverage factor. We propose that this can be achieved by combining the KDD (Bottom-Up Approach) and the Business-Driven (Top-Down approach). The first approach covers all existing dataset (which generates new business opportunities (GBO)), while the second will ensure building the personalized profiles by the business units based on a market research. In addition, we propose to add the time dimension to segmentation. Just like the behavior and Socio-demographic segmentation answer the question of who will buy and where, adding the time dimension answers the question when. This leads to more personalized understanding of the next best activities for each customer.

Personalization Framework Architecture

A predefined dataset for the business considered is the basic input to the framework. In banking for example, the dataset is of customer demographics, customer account information and transactions. This data is routinely saved in banks and is extracted from operational source systems. The proposed steps of the personalization framework are as follows:

- 1) Behavior-Based segmentation: all transaction done by customers will be captured from the operational sources to be segmented by behavior. The resulting segmentation (grouping of customers by spending behavior) will be the input to the next step. The Input KPIs are effective statistical measures that have the ability to provide effective understanding of the customers' behavior. The KPIs will be calculated for the segments resulting from step 1 and will be the input for the data reduction in step 2. The KPIs calculated are: average number and amount of purchase, percentage of spend for each merchant category and number of different merchants used.
- 2) Data reduction using PCA or factor analysis. PCA is used for reducing the dimensionality and generates some simple, clear and fewer input components for the clustering algorithm in the next step.
- 3) Identification of the Segments by Cluster Modeling: Clustering models are unsupervised techniques, appropriate when there is no target output field. They analyze the input components and group records with respect to the identified input usage patterns. The output will be different clusters each one has the same behavior.

More than one dimension can be added before the profiling step to enrich the meaning and functionality of the segment like RFM segment (Recency, Frequency, and Monetary), demographics (Considered in our case study) and Customer value. Considering a hybrid dimensions gives a better view for the personalized needs by different criteria. Applying the Demographics association to the output clusters will define the relation and distribution of the demographical parameters through each segment. Now the criteria

will define the case in more personalized analysis, so the action will be taken in the right time to the right segment.

- 4) Evaluation and Profiling of the Revealed Segments: This step seeks guidance from the marketers for selecting the most effective segmentation. Tasks in this phase include profiling and labeling of the revealed segments (Business Driven). Profiling supports the business evaluation of the segments as well as the subsequent development of effective marketing strategies tailored for each segment. The marketing research information is used to evaluate and enrich the behavioral segments. The segments will be labeled based on their identified profiles, for instance "Business Travelers," instead of "Segment 1."

Profiling with supervised (classification) models is considered as a modification for the framework to increase the automation process of the profiling step. Decision trees in particular, due to the intuitive format of their results, are typically used to outline the segment profiles.

- 5) Segment groups by time dimension: The power of time dimension as an analysis factor had been neglected by many researches. The technique used here is un-predictive association rule technique. For example, the travelers group will have highest activity in summer season, so targeting the same segment in another season will be miss leading as a targeting plan.

The output will be a group of a personalized understandable segments dynamically changed according to their operational transactions log captured from sources and automatically profiled by the classification technique. The output will be the base for any CRM next activities in a personalized handling. (Business plans to be applied).

4. Case Study: Credit Cards holders' Personalization

In this section we present a theoretical case study that follows 5 bank customers, out of the total training data set, over the framework architecture steps to illustrate the working of the proposed framework. We are considering a case study in the banking sector, where the bank's objective is to reveal the underlying customer groups with similar usage characteristics in order to design differentiated marketing strategies. Their primary goal is to develop new card types and reward programs tailored to the different customer characteristics.

4.1 Dataset

The data set covers aspects of credit card use including number and amount of purchases, cash advance transactions, payments as well as customer demographics, and account details. Purchase transactions in particular were recorded per merchant code, and track the purchasing habits of each customer. The data set consists of 50,000 credit cards customers, we considered 5 customers as a sample, and their transactions, customer's demographics and accounts information are presented in Tables 1, 2, 3. The predefined data set of customers and their dimensions needed to be identified, modeled, cleansed, clustered, associated and profiled as per the framework architecture described in Section3.

Table 1 Customer Demographics

CUST_ID	JOB	GENDER	RESIDENCE	LONGEVITY	AGE_BRACKET
00A	Engineer	FEMALE	CAIRO	2 Years	31-40
00B	Doctor	MALE	ALEX	3 Years	21-30
00C	Officer	MALE	ASSIUT	6 Months	21-30
00D	Nurse	FEMALE	CAIRO	13 Months	31-40
00F	Business Man	MALE	MANSORA	15 Months	41-50

Table 2 Account Information

ACC_ID	OPEN_DATE	CREDIT_LIMIT	STATUS	ACCOUNT_TYPE	PRODUCT
001	JAN2014	25,000	Normal	Main	Visa Gold
002	OCT2013	15,000	Normal	Additional	Master Classic
003	JUL2015	80,000	Dormant	VIP	Visa Platinum
004	AUG 2014	20,000	Normal	Main	Visa Classic
005	DEC 2013	90,000	Delinquent	VIP	Master Platinum

Table 3 Transaction Information

ACC_ID	TRX_ID	TRX_TYPE	MCC	CUR	COUNTRY	Localization	MONTH	AMOUNT
001	T_001	POS	Shoes	EGP	EGYPT	LOCAL	JUL	3000
001	T_002	POS	Clothing	EUR	FRANCE	INTER	AUG	2000
001	T_003	POS	Jewelry	EGP	EGYPT	LOCAL	JUL	3000
002	T_004	POS	air tickets	EUR	FRANCE	INTER	AUG	2000
002	T_005	INTERNET	hotel bookings	USD	USA	INTER	SEP	1500
002	T_006	POS	Travel agencies	EGP	EGYPT	LOCAL	JUL	3000
003	T_007	POS	Restaurants	EUR	FRANCE	INTER	AUG	2000
003	T_008	POS	Clubs	USD	USA	INTER	SEP	1500
004	T_009	POS	Cosmetics	USD	USA	INTER	SEP	1500
004	T_010	POS	Air tickets	EUR	DUBAI	INTER	SEP	2500
005	T_01	POS	car rentals	USD	USA	INTER	SEP	1500

- Customer Demographics: Each record describes the baseline segmentation for the customer as presented in Table 1,
- Accounts Information: Each record describes the account information as presented in Table 2,
- Transactions Information: Each record describes the transactions information as presented in Table 3

4.2 MCMD Framework Applied

Behavior-based Segmentation: Since the bank's primary objective was to differentiate customers according to their general purchasing behaviors, a high-level categorization into 4 merchant categories; TRAVEL (Travel agencies/ air tickets/ hotel bookings/ car rentals etc.), ACCESSORIES (shoes/ jewelry/ cosmetics), ENTERTAINMENT (restaurants/ coffee houses/ bars/ clubs etc.), and APPAREL (clothing stores)

- 1) Sample KPIs of Credit Card Usage • Frequency of purchases and cash advance • Recency of last purchase (time since the last purchase) • Monthly average number and amount of purchase • Monthly average balances • Number of different merchant categories with purchases.

The analytical process that was followed included three main steps.

- **Extraction** of the segmentation dimensions by using a supervised data reduction technique.

- **Identification** of the customer segments through the application of a clustering model.

- **Profiling** of the segments.

2) Applying the data reduction technique:

As their name implies, data reduction techniques aim to effectively reduce the data's dimensions and removing redundant information. They do so by replacing the initial set of fields with a core set of compound measures which simplify subsequent

modeling while retaining most of the information of the original attributes.

Table 4 Accounts Association to each input KPI

ACC_ID	%ACCESSORIES	%APPAREL	%TRAVEL	%ENTERTAINMENT
001	66.6	33.3	0	0
002	0	0	100	0
003	0	0	0	100
004	50	0	50	0
005	0	0	100	0

Table 5 Pairwise correlation coefficients among the original input fields

	%ACCESSORIES	%APPAREL	%TRAVEL	%ENTERTAINMENT
%ACCESSORIES	1.00	0.69	0.69	0.00
%APPAREL	0.69	1.00	0.00	0.00
%TRAVEL	0.69	0.00	1.00	0.00
%ENTERTAINMENT	0.00	0.00	0.00	1.00

- I) Relate each customer to the most associated merchant to simulate the spending behavior as seen in table 4
- II) Perform the pairwise Pearson correlation coefficients among the inputs. As shown in table 5, there are some significant correlations among specific usage fields.
- III) The 4 fields of the percentage of purchases by merchant type were used as inputs to a PCA (Principle Components Analysis) model which extracted a set of 2 components, based on the "Eigenvalues over 1" criterion and a Varimax rotation as explained in table 6.

Eigenvalues can also be expressed in terms of a percentage of the total variance of the original fields. The eigenvalue of each component is the sum of squared loadings (correlations) across all input fields. Thus, each eigenvalue denotes the total variance or total information interpreted by the respective component.

Eigenvalues = (Record value – mean value of field)/standard deviation of the field

Table 6 Varimax rotation

Total variance explained			
Component	Eigenvalue	Percentage of variance	Cumulative %
1	1.52	45.2	45.2
2	1.24	27.2	72.4
3	.74	18.3	90.7
4	.36	9.3	100.0

- IV) To understand the meaning of each component, perform a correlation among the derived components and the original inputs. Rotation is a recommended technique to apply in order to facilitate the interpretation process of the components. Rotation minimizes the number of fields that are strongly correlated with many components and attempts to associate each input to one component as shown in table 7.

Table 7 the rotated component matrix

	Component 1	Component 2
%ACCESSORIES	0.97	0.00
%APPAREL	0.85	0.00
%TRAVEL	0.00	0.95
%ENTERTAINMENT	0.00	0.86

- V) Interpretation of the extracted principal components

Table 8 Interpretation of the extracted principal components

Component	Features
Component 1	Fashion purchases Component 1 represents fashion purchases. Purchase of clothes and accessories are correlated with each other and with component 1.
Component 2	Component 2 represents use of the card to cover traveling expenses Spending on entertainment ,

- 3) **Clustering:** Clustering models are unsupervised techniques, appropriate when there is no target output field. They analyze a set of inputs and group records with respect to the identified input data patterns. The major advantage of the clustering techniques is that they can efficiently manage a large number of attributes and create data-driven segments.

TwoStep cluster will be applied: As its name implies, this scalable and efficient clustering model, processes records in two steps. It suggests a clustering solution by automatic clustering: the optimal number of clusters can be automatically determined by the algorithm according to specific criteria. It can efficiently handle large datasets. It incorporates a special probabilistic distance measure that can efficiently accommodate both continuous and categorical input attributes. The clustering algorithm runs to produce a recommended strong 2 clusters.

The percentage of dataset distribution over clusters; Cluster1 includes 60.3% from credit card holders tends to buy clothing, Jewelry, and accessories, and Cluster2 includes 39.7% shows increased usage for traveling expenses and entertainment.

Socio-Demographical dimension

Demographical is another dimension to be added for increase the level of personalization as it uses the inputs express about the social, culture and nature of the customers.

Table 9 Demographical profile for each cluster

		Cluster 1	Cluster 2	Total
Gender (%)	Male	30	70	60
	Female	70	30	40
Age (%)	21–30	0	90	20
	31–40	90	0	40
	41–50	10	10	40

So, we can see cluster 1 concern a larger number of middle age and female customers, otherwise the cluster 2 concern the younger or older customers and mostly the male customers.

4) Segments Profiling (Start Top-Down Approach)

This process demands the close co-operation of data miners and marketers in order to fully interpret the derived clusters and assess their business value, their usability, and action-ability before deploying the derived segmentation. By combining the behavioral segmentation and demographics as a sample of 2 dimensions we can provide the business units by new patterns could be studied and profiled as follows.

Segment 1 – Fashion Ladies

Behavioral Profile

- Increased purchases at apparel stores
- Increased purchases at accessory stores (jewelry, shoes, etc.)

Segment Size: 40%

Contribution to Overall Purchase Amount 40.5%

Demographic Profile: consists of middle-aged women.

Segment 2 – Travelers and Fun Lovers

Behavioral Profile

- Increased purchases for entertainment (restaurants, coffee houses, etc.)
- Use of card mainly for traveling and vacations (hotel bookings/air tickets/car rentals, etc.)
- The highest international usage with a large percentage of the total purchases made abroad
- Increased purchases over the Internet (for e-booking)
- Increased total amount of purchases (tickets are expensive)

Segment Size: 60%

Contribution to Overall Purchase Amount: 59.5%

Demographic Profile: Mostly young customers

- 5) **Time Series Analysis dimension:** Adding the time dimension proved to be an effective factor to consider in segmentation as it identifies the time of the year or the time of the day when an activity tends to increase to predict the next best activity of the segment. The Association between the segments and their activities through months of the year presented in table 10

So, the preferred targeting campaigns for the fashion ladies will be more tailored in summer season especially in July to be titled by a summer fashion lady, and for the Travelers and fun lovers also will be in summer but the more targeting month will be September (They can be a summer travelers or winter travelers based on association)

Table 10 Segments to time series association

	Month				
	JAN to JUN (%)	JUL (%)	AUG (%)	SEP (%)	OCT to DEC (%)
Fashion Ladies	0	50	25	25	0
Travelers & Fun Lovers	0	14.5	28.5	57	0

Classification for auto profiling

To enhance the automation of groups profiling and decrease the business effort and time we will apply a classification technique (decision tree) to auto profile the inputs, with rules extracted from the first profiling run. The rules displayed in table 11

Table 11 Extracted rules for association algorithm

	Rule 1	Rule 2	Rule 3	Rule 4
Fashion Ladies	purchases at apparel stores	purchases at accessory stores	Middle-age	Gender is women
Travelers & Fun Lovers	purchases for entertainment	purchase for traveling and vacations	Younger age	Gender is men

Business Plans

The segmentation results were also used in new product development and the offer of customized products and reward programs. The selected strategy involved offering specialized types of cards, with suitable rewards covering the differentiated needs of customers.

A travel-oriented card was the bank's offer to the “**Travelers and Fun lovers**” segment. Partnerships with specific airlines, travel agencies, hotels, and car rental companies made it possible to offer specific benefits to frequent travelers. For instance, customers could take advantage of the card's reward program and use their cards to collect **air miles** or get a special booking discount at a partnering hotel. The card's benefits also included the offer of travel insurance when traveling abroad. Additionally, the bank started to address these customers through its subsidiary travel agency and to send them travel offers on a regular basis. All customers began to receive a leaflet on travel bargains and the card was further advertised at airports, on travel Internet sites, and in magazines.

A “**fashion**” card with a relevant reward scheme was the bank's proposition for the “Fashion ladies” of segment 6. The rewards tied to card usage included rebates and zero-rate installments for purchases at retail affiliates. This card was advertised in women's fashion magazines. Stands with card brochures were also positioned in affiliated fashion stores. A bargain fashion catalogue was also mailed to members of this segment, promoting card purchases at partnering stores.

5. Conclusion and Future Work

In customer-centric businesses, improving the personalization aims to provide a better understanding of the customer and to support the identification of new marketing opportunities through design and development of new products/services tailored to each segment's characteristics. The tailored marketing strategies ultimately increase the level of customer satisfaction and their loyalty and increase profitability.

The proposed MDMC personalization framework merges machine learning techniques with traditional personalization strategies and aims to enhance on the previous personalization approaches' drawbacks. Mainly it aims to improve the automation of profiling and to keep the cost low and the CRM processes' speed high. This is done by applying an unsupervised clustering technique in knowledge data discovery step followed by a supervised classification technique that gets its rules from the first run business profiling. The automation leads to controlling the costs related to marketing actions, and avoids leaving too much power to front office personnel.

Satisfying a high level of targeting measure and keeping the coverage factor fulfilled are both important, but are often conflicting and come at the expense of one another. In the framework proposed, we combined the KDD (Bottom-Up Approach) and the Business-Driven (Top-Down approach) to be able to balance both coverage and targeting. This is achieved as the first approach is applied first to covers all existing dataset, and the second approach is then applied to profile each customer based on a market research.

To further enhance the level of targeting we proposed a new dimension to be included in the framework, which is the time dimension that increases the level of targeting based on specific time or season as per nature of need.

Future work to enhance the personalization accuracy is to consider additional unusual effective dimensions that can affect the personalization results like the political and economic sudden changes. Automation of the action to be delivered after the framework run is another large area for future work in this area. Our own future work will focus on applying the framework architecture on a real life banking dataset to measure the results in actual marketing campaigns and compare the results with results from previous approaches.

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